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10. In a hexaploid wheat(2n = 6 x = 42) the haploid (n) and the basic(x) number of chromosomes respectively are

a. n = 7 and x = 21 b. n = 21 and x = 21
c. n = 7 and x = 7 d. n = 21 and x = 7

II Fill in the blanks

1. Economically important crop plants with superior quality are raised by_____.
2. A protein rich wheat variety is _____.
3. _____is the chemical used for doubling the chromosomes.
4. The scientific process which produces crop plants enriched with desirable nutrients is called _____.
5. Rice normally grows well in alluvial soil, but _____ is a rice variety produced by mutation breeding that grows well in saline soil.
6. _____ technique made it possible to genetically engineer living organism.
7. Restriction endonucleases cut the DNA molecule at specific positions known as _____.
8. Similar DNA fingerprinting is obtained for _____.
9. _____ cells are undifferentiated mass of cells.
10. In gene cloning the DNA of interest is integrated in a _____.

III State whether true or false. If false, write the correct statement

1. *Raphano brassica* is a man-made tetraploid produced by colchicine treatment.
2. The process of producing an organism with more than two sets of chromosome is called mutation.
3. A group of plants produced from a single plant through vegetative or asexual reproduction are called a pureline.
4. Iron fortified rice variety determines the protein quality of the cultivated plant
5. Golden rice is a hybrid.
6. Bt gene from bacteria can kill insects.
7. *In vitro fertilisation* means the fertilisation done inside the body.
8. DNA fingerprinting technique was developed by Alec Jeffrey.
9. Molecular scissors refers to DNA ligases.

IV Match the following

	Column A	Column B
1.	Sonalika	*Phaseolus mungo*
2.	IR 8	Sugarcane
3.	Saccharum	Semi-dwarf wheat
4.	Mung No. 1	Ground nut
5.	TMV – 2	Semi-dwarf Rice
6.	Insulin	*Bacillus thuringienesis*
7.	Bt toxin	Beta carotene
8.	Golden rice	first hormone produced using rDNA technique

V Understand the assertion statement, justify the reason given and choose the correct choice

a. Assertion is correct and reason is wrong
b. Reason is correct and the assertion is wrong
c. Both assertion and reason is correct
d. Both assertion and reason is wrong.

1. **Assertion:** Hybrid is superior than either of its parents.
Reason: Hybrid vigour is lost upon inbreeding.
2. **Assertion:** Colchicine reduces the chromosome number.
Reason: It promotes the movement of sister chromatids to the opposite poles.
3. **Assertion:** rDNA is superior over hybridisation techniques.

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